

Dr. Eric Janusson

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Professional Summary

PhD Analytical Chemist and Scientific Software Developer specializing in analytical method development and computational analytics. Expertise spans mass spectrometry (QTOF, TWIMS, Orbitrap, ICP-MS), open-source software for multidimensional analysis, and analytical instrumentation (HV/UHV systems, ion sources). Proven track record in regulated laboratory operations (ISO/IEC 17025, Health Canada Section 56), 5N+ purity certification, and scalable R&D.

Core Competencies

- **Software Development:** Python (pandas, polars, matchms, pyteomics), spectral-similarity algorithms, automated databasing, interactive data visualization/web development (React, D3.js, MDX), version controlled, and config-driven workflows.
- **Mass Spectrometry & Chemical Separations:** ESI, EI, LC-MS/MS, GC-MS, IC, TWIMS; Orbitrap, QTOF, QqQ, ICP-MS, and magnetic-sector analyzers.
- **Analytical Instrumentation:** Analytical facility commissioning; HV/UHV systems, ion sources, ion optics, gas manifolds, and board-level electronics.
- **Regulated R&D Leadership:** ISO/IEC 17025 and cGMP-aligned laboratory operations, 5N+ purity certification, capital procurement, Health Canada Section 56 exemptions, and chain-of-custody controls.

Professional Experience

Independent Software Developer | North Vancouver, BC | January 2026 – *Present*

- Released MassFlow, an open-source platform for reproducible, vendor-agnostic MS/MS annotation and spectral analysis.
- Developed PySharpe, a modular open-source web application for portfolio optimization, backtesting, and risk analysis.

Research Associate / Acting Technical Lead | TMC Manufacturing | North Vancouver, BC | 2024 – 2026

- Managed industrial-scale ultra-high-purity stable isotope production, including enrichment systems, custom ion sources and optics, UHV systems, gas manifolds, and high-voltage electronics.
- Designed and validated trace-element methods for $\geq 99.999\%$ (5N+) purity certification of rare-earth and inorganic materials, replacing external analysis with an automated ISO/IEC 17025-aligned in-house platform and reducing turnaround from weeks to hours.
- Developed and validated a bespoke potentiometric chloride assay with sub-ppm detection limits, reducing halogen contamination by 99.997% and effectively eliminating out-of-specification batches.
- Led \$515K CAD in capital procurement for ICP-MS and IC instrumentation, authoring the business case and bringing critical purity verification in-house.
- Developed a historical analytical database for batch trending, root-cause investigation, and quantitative uncertainty analysis.
- Supervised and mentored cross-functional chemistry and engineering staff, directly contributing to the promotion of two junior scientists.

Lead Chemist | Delic Labs | Vancouver, BC | 2021 – 2022

- Commissioned a federally authorized analytical facility from the ground up, successfully installing and qualifying high-resolution LC-MS, GC-MS, HPLC, and spectroscopic platforms.
- Authored Standard Operating Procedures (SOPs) and defined an analytical services framework fully aligned with ISO/IEC 17025.
- Engineered proprietary Python pipelines for untargeted MS profiling and molecular networking, annotating >9,500 spectra and identifying >50 previously unreported metabolites in complex botanical matrices.
- Secured Health Canada Section 56 exemptions for Schedule I controlled substances, ensuring audit-ready inventory and custody controls.

Postdoctoral Research Associate (Cronin Group) | University of Glasgow | Glasgow, UK | 2017 – 2021

- Built automated Python data pipelines to process and interpret high-dimensional LC-TWIMS-HRMS datasets, mentoring graduate researchers in computational analytics.
- Designed LC-TWIMS-MS analytical methods, integrated with a robotic synthetic platform, to resolve the real-time self-assembly mechanisms of inorganic polyoxometalate (POM) nanomaterials.

- Discovered and published a novel carbohydrate-directed synthetic route conferring isomer and conformational control over self-assembling POM clusters.

PhD Researcher & Graduate Laboratory Instructor | University of Victoria | Victoria, BC | 2012 – 2017

- Developed real-time ESI-MS/MS (QTOF, QqQ) methodologies to detect, characterize, and monitor transient organometallic intermediates in Pd-catalyzed cross-coupling reactions.
- Published studies on electrospray ionization (ESI) ion-source physics, ion pairing, solvent effects, and charge-tagging strategies for selective mass spectrometric analysis.
- Supervised senior undergraduates through independent, publication-level research.

Education

- **PhD, Analytical Chemistry** | University of Victoria, BC (2017)
 - *Dissertation*: "Development of mass spectrometry techniques for real-time reaction monitoring."
- **BSc, Chemistry** | University of Victoria, BC (2012)

Certifications

- Agilent LC-QTOF, GC-MS, FTIR, and UV-Vis Operator Training & Certification (2021)
- Waters Synapt G2 TWIMS & QTOF Operator Training & Certification (2018)

Selected Publications

- Janusson, E. et al. "Synthesis of polyoxometalate clusters using carbohydrates as reducing agents leads to isomer-selection." *Chem. Commun.*, 2019.
- Janusson, E. et al. "Step-by-step real-time monitoring of a catalytic amination reaction." *Chem. Commun.*, 2019.
- Janusson, E. et al. "Real-time analysis of Pd₂(dba)₃ activation by phosphine ligands." *Chem. Commun.*, 2017.
- Janusson, E. et al. "Selective mass spectrometric analysis of thiols using charge-tagged disulfides." *Analyst*, 2016.

Full list of publications available at <https://orcid.org/0000-0002-3207-7067>